



Te Pūkenga

- School of Innovation, Design & Technology -

IT5506 Bachelor of Information Technology IT5487 Diploma in Information Systems (Level 5)

Practical Lab 1.1

Configure switch-based Ethernet network

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Score: **_____ / 100 points**

Details

Due Date: **18 August 2023**

This Lab is worth 10% of the over-all course grade.

Learning Outcomes:

On successful completion of this assessment, the learner will be able to:

1. Describe network protocol models and devices to explain the layers of communications in data networks.
2. Design and calculate IP addresses and subnet masks for both IPv4 and IPv6 for given simple networks, using IPv4 and IPv6.
3. Explain fundamental Ethernet concepts.

Task Overview

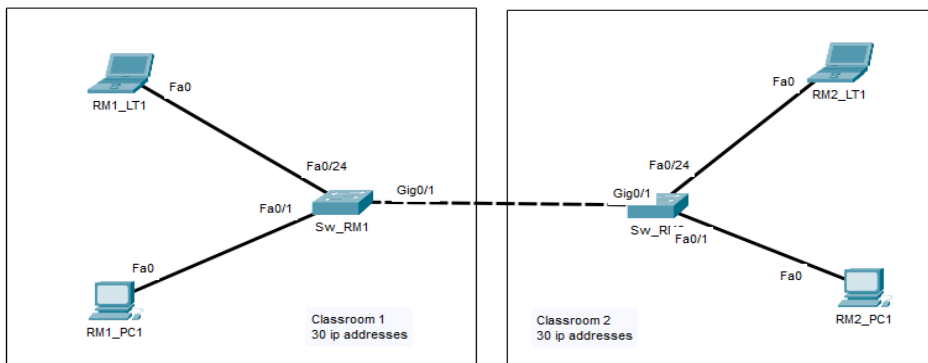
The assessment is made up of 2 files that will need to be completed and uploaded to Moodle for marking. You can find them on our class Moodle page under the **Assessment Details** tab

1. Lab 1.1 assessment tasks v2023T2 (this file)
2. Lab 1.1 PT assessment file v2023T2

For this lab, you will be using the Cisco Packet Tracer application software to configure a small switch-based Ethernet network consisting of 2 ethernet LAN switches and 2 personal computers and 2 laptops as shown in the topology diagram below.

Both desktop computers have been preconfigured with IP addresses. You will need to add the 2 laptops, using appropriate media to connect them to the switches and then configure the IP address settings. Network switches will need to be connected using the appropriate network media then configured based on network specifications.

Topology Diagram



IP Addressing Table

The network will use the IP Address range of **10.10.10.0** and subnet mask of **255.255.255.0**

Device	Hostname	IP Address	Subnet Mask	Default Gateway
Switch	Sw_RM1	10.10.10.253	255.255.255.0	
Switch	Sw_RM2	10.10.10.252	255.255.255.0	
Laptop	RM1_LT1	10.10.10.30	255.255.255.0	10.10.10.254
Laptop	RM2_LT1	10.10.10.60	255.255.255.0	10.10.10.254
Computer	RM1_PC1	10.10.10.1	255.255.255.0	10.10.10.254
Computer	RM2_PC1	10.10.10.31	255.255.255.0	10.10.10.254

INSTRUCTIONS

Part One: Create Network Topology

Task 1: Download the Packet Tracer assessment file

Download the **Lab 1.1 PT assessment file v2023T2** from our class Moodle page. You will need Packet Tracer application version 8.1 or higher

Note. To download a copy of Packet Tracer v8.1.1 64-bit application you can find a download on the Cisco Network Academy Program (CNAP) website at www.netacad.com

Task 2: Build the network based on the Topology Diagram

9 marks

1. Based on the topology diagram, add the appropriate devices needed to build the LAN i.e two laptops.
 - a. The computers have been connected to the switches and preconfigured with their IP address settings.
2. Based on the TIA/EIA network media standards select the appropriate network patch cables and connect the laptops to the specified switch ports. You will then connect the switches to each other.
 - a. Connect switch Sw_RM1 port g0/1 to switch Sw_RM2 port g0/1. What type of network patch cable did you use?

Copper Cross-over

- b. Connect Laptop RM1_LT1 to switch Sw_RM1 port f0/24. What type of network patch cable did you use?

Copper Straight-Through

- c. Connect Laptop RM2_LT1 to switch Sw_RM2 to port f0/24. What type of network patch cable did you use?

Copper Straight-Through

Part Two Configure Network Devices

Each classroom needs 30 IP addresses to cater for all the computing devices needed in the classroom. Classroom 1 will get the first set of 30 IP addresses. Classroom 2 will get the second set of 30 IP addresses. The third set of 30 IP addresses has been reserved for use by the school printers, servers and other networking devices.

Task 3: Configure end-user devices

16 marks

Configure the laptop IP addresses as specified in the IP Addressing table values. Note the computers have been preconfigured with IP addresses.

1. Based on the IP Addresses table what is the overall Network IP address

10.10.10

2. What is the range of valid IP addresses needed for classroom 1.

10.10.10.1 – 10.10.10.30

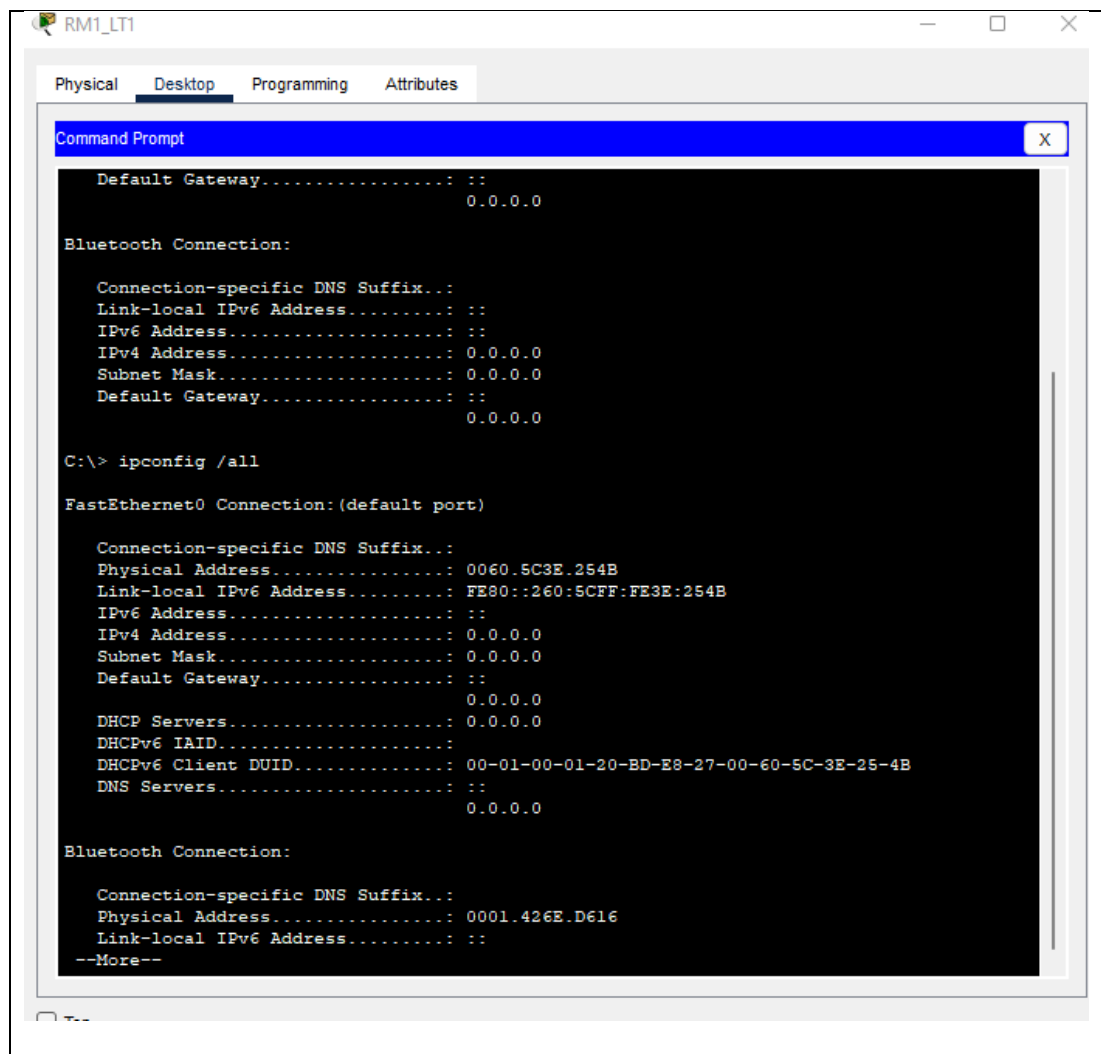
3. What is the range of valid IP addresses needed for classroom 2

10.10.10.31 – 10.10.10.60

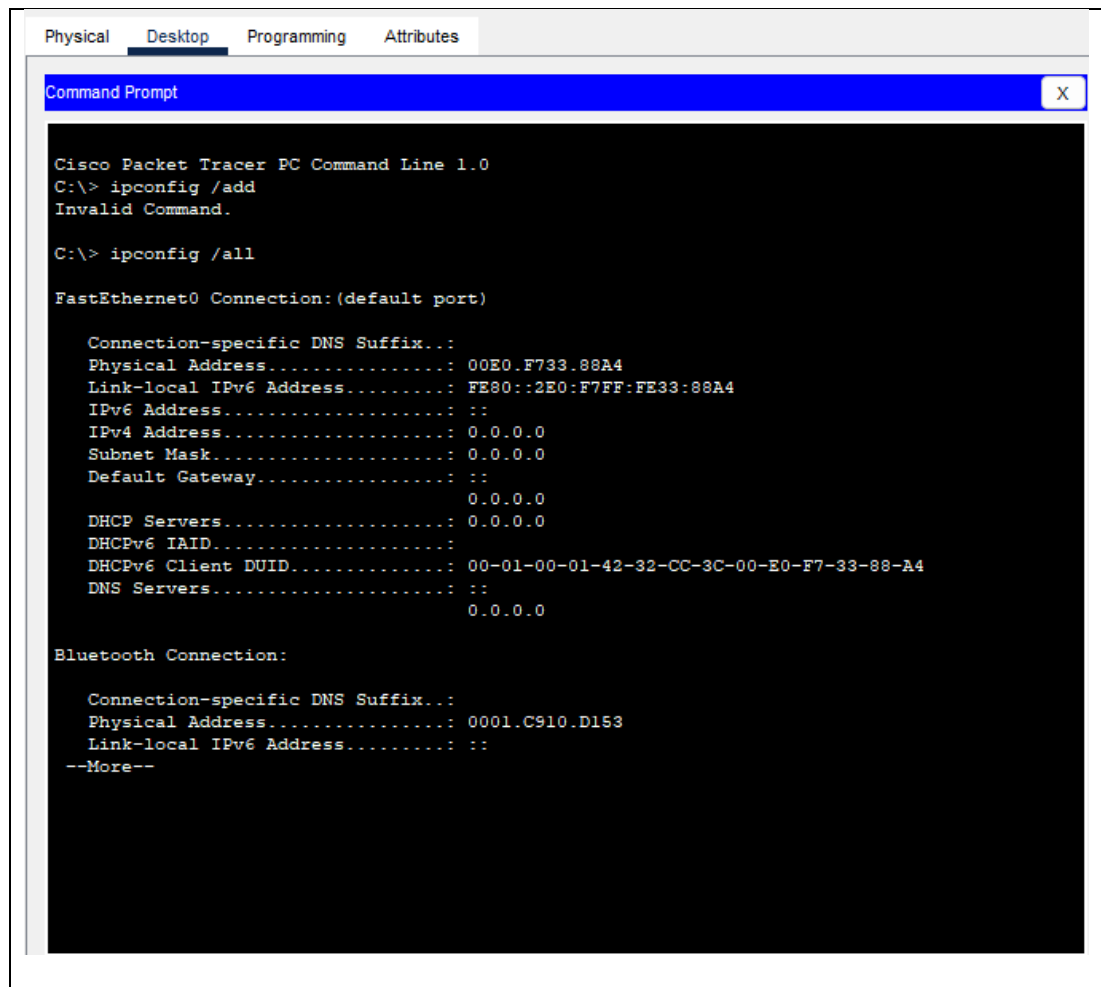
4. What is the range of valid IP addresses needed for the servers and printers

10.10.10.61-10.10.10.253

5. Configure laptop RM1_LT1 IP address, subnet mask and default gateway settings.
 - a. Open the command prompt and issue the **ipconfig /all** command. Take a screenshot of the output and paste image below



6. Configure laptop RM2_LT1 IP address, subnet mask and default gateway settings.
 - a. Open the command prompt and issue the **ipconfig /all** command. Take a screenshot of the output and paste image below



Task 4: Configure network switches

34 marks

To complete this task, you need to access the switches management interface to be able to configure system settings. To do this you must connect the laptop to the switch console port and use the default Packet Tracer terminal emulation application called **Terminal**.

1. Connect the laptops to the switches console management port. According to TIA/EIA wiring standards what type of network patch cable did you use.

Console cable

2. Open the Packet Tracer terminal emulation program.
 - a. Click on your laptop
 - b. Click on Desktop tab
 - c. Click on the **Terminal** application
 - d. Configure the following console management access settings

Bits per second	9600
Data bits	8
Parity	None
Stop bits	1
Flow control	None

- e. Click on OK button.
- f. The terminal application user interface screen opens.

3. Configure the following switch settings

Setting	Switch0	Switch1
Host name	Sw_RM1	Sw_RM2
Secret Password	itsasecret	itsasecret
Console password	letmein	letmein
VTY password	cisco	cisco
Encrypt all passwords	encrypted	encrypted
Banner message	Authorised Access Only	Authorised Access Only
VLAN IP address	10.10.10.253	10.10.10.252
Save the settings	saved	saved

4. What command is used to show the saved configuration file settings.

show startup-config

5. Name 2 commands used to show the IP address settings on a switches interface

show running-config, show ip interfaces

Part Three: Testing and Verification

Task 5: Test connectivity between network devices

17 marks

1. From laptop Rm1_LT1 open the command prompt
 - a. Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful.

RM1_PC1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
No	Yes	Yes	No	No

- b. Issue the **arp -a** command to view the layer 2 MAC addresses of devices that have successfully connected to the laptop. Take a screenshot of the command output

<pre>C:\>arp -a Internet Address Physical Address Type 10.10.10.252 0090.0c1d.b80e dynamic 10.10.10.253 0004.9a81.b7ce dynamic</pre>		
---	--	--

2. From laptop RM2_LP1 open the command prompt
 - a. Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful.

RM1_PC1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
No	Yes	Yes	Yes	No

- b. Issue the **arp -a** command to view layer 2 MAC addresses of devices that have successfully connected to the laptop. Take a screenshot of the command output

<pre>C:\>arp -a Internet Address Physical Address Type 10.10.10.252 0090.0c1d.b80e dynamic 10.10.10.253 0004.9a81.b7ce dynamic</pre>		
---	--	--

3. From switch Sw_RM1 open the cli interface
 - a. Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful.

RM1_PC1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
No	Yes	Yes	Yes	No

- b. Issue the **show mac-address table** command to view the switches addressing table entries. Take a screenshot of the command output.

--


```
Sw_RM1# show mac-address
```

Mac Address Table			
Vlan	Mac Address	Type	Ports
1	0006.2a74.0719	DYNAMIC	Gig0/1
1	0090.0c1d.b80e	DYNAMIC	Gig0/1

4. From switch Sw_RM2 open the cli interface
- Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful.

RM1_PC1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
No	Yes	Yes	Yes	No

- Issue the **show mac-address table** command to view the switches addressing table entries. Take a screenshot of the command output

```
Sw_RM2#show mac-address
```

Mac Address Table			
Vlan	Mac Address	Type	Ports
1	0004.9a77.e619	DYNAMIC	Gig0/1
1	00e0.f733.88a4	DYNAMIC	Fa0/24

5. Explain why the switches addressing table only contains layer two MAC addresses.

Switches addressing table only contains layer two mac address because it makes forwarding decisions. This is to determine how to handle incoming frames.

Part Four: Troubleshooting

Task 6: Fix the incorrect configuration settings

8 marks

Based on findings from connectivity tests both desktop computers cannot be reached by the laptops and the switches. You must check the media type and address configuration settings of the computers and identify the reason why connectivity tests failed.

1. Investigate computer RM1_PC1 settings
 - a. Identify the faults

IP address, subnet mask and default gateway are configured incorrectly

- b. Explain how you fixed the faults.

Use addressing table to configure correctly

2. Investigate computer RM2_PC1 settings and identify the faults
 - a. Identify the faults

The PC was turned off and incorrect cable connection to the switch

- b. Explain how you fixed the faults

Options, preferences and untick hide physical tab and hide all devices tab. Click on RM2_PC1 and turn PC on showing green light. Change cable to copper straight through connection.

Task 7: Verify faults have been fixed

16 marks

Once you have completed making the fixes conduct connectivity tests on the computers

1. On computer Rm1_PC1 open the command prompt.
 - a. Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful

RM1_LT1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
Yes	Yes	Yes	Yes	Yes

- b. Issue the **arp -a** command to view layer 2 MAC addresses of devices that have successfully connected to the laptop. Take a screenshot of the command output.

```
C:\>arp -a
Internet Address      Physical Address      Type
10.10.10.31           0001.43de.e979        dynamic
10.10.10.60           00e0.f733.88a4        dynamic
10.10.10.252          0090.0c1d.b80e        dynamic
10.10.10.253          0004.9a81.b7ce        dynamic
```

- c. Issue the **ipconfig /all** command to view the computers ip address settings. Take a screenshot of the command output


```

C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 00E0.F733.88A4
    Link-local IPv6 Address.....: FE80::2E0:F7FF:FE33:88A4
    IPv6 Address.....: ::
    IPv4 Address.....: 10.10.10.60
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        10.10.10.254
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-42-32-CC-3C-00-E0-F7-33-88-A4
    DNS Servers.....: ::
                        0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0001.C910.D153
    Link-local IPv6 Address.....: ::
    --More--

```

3. From switch Sw_RM1 open the cli interface.
 - a. Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful.

RM1_PC1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
Yes	Yes	Yes	Yes	Yes

- b. Issue the **show mac-address table** command to view the switches addressing table entries. Take a screenshot of the command output

```

Sw_RM1# show mac-address
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.42de.e979    DYNAMIC     Gig0/1
1       0006.2a74.0719    DYNAMIC     Gig0/1
1       0060.5c3e.254b    DYNAMIC     Fa0/24
1       0090.0c1d.b80e    DYNAMIC     Gig0/1
1       00e0.f733.88a4    DYNAMIC     Gig0/1

```

4. From switch Sw_RM2 open the cli interface
 - a. Use the **ping** command to test connectivity with all other device IP addresses. Enter **yes** or **no** to indicate the connectivity test was successful

RM1_PC1	Sw_RM1	Sw_RM2	RM2_LT1	RM2_PC1
Yes	Yes	Yes	Yes	Yes

- b. Issue the **show mac-address table** command to view the switches addressing table entries. Take a screenshot of the command output

```

Sw_RM2# show mac-address
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.42de.e979    DYNAMIC     Fa0/1
1       0004.9a77.e619    DYNAMIC     Gig0/1
1       0004.9a81.b7ce    DYNAMIC     Gig0/1
1       0060.5c3e.254b    DYNAMIC     Gig0/1
1       00e0.f733.88a4    DYNAMIC     Fa0/24
-

```

Assessment complete, upload this file and your Packet Tracer file to Moodle

Marking Schedule

Tasks	Mark
Task 2 Build the network based on the Topology Diagram	9
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b. Connect laptop to switch and media used	3
c. Connect laptop to switch and media used	3
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c. Class 2 ip address range	2
d. School servers and printers ip address range	2
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b. Laptop RM1_LT1 arp screenshot	1
c. Laptop RM2_LT1 ping test	2.5
d. Laptop RM2_LT1 arp screenshot	1
e. Switch Sw_RM1 ping test	2.5
f. Show mac-address table screenshot	1
g. Switch Sw_RM1 ping test	2.5
h. Show mac-address table screenshot	1
i. Why does a switch's address table only contain layer 2 mac addresses	3
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b. Fix faults	2
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b. Computer RM1_PC1 ping test, arp and ipconfig screenshot	4.5
c. Switch Sw_RM1 ping test and mac-address table screenshot	3.5
d. Switch Sw_RM1 ping test and mac-address table screenshot	3.5
Total	100